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Subject: Programming with Java	
ASSIGNMENT No : 5	
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

```
import java.util.ArrayList;

import java.util.Vector;


public class CollectionDemo {


    public static void main(String args[]) {


        ArrayList<String> fruits = new ArrayList<>();

        fruits.add("Apple");

        fruits.add("Banana");

        fruits.add("Mango");

        System.out.println("ArrayList after adding: " + fruits);


        fruits.remove("Banana");
```

```
System.out.println("ArrayList after removing Banana: " + fruits);

fruits.add(1, "Grapes");
System.out.println("ArrayList after inserting Grapes: " + fruits);

System.out.println("Element at index 0: " + fruits.get(0));
System.out.println("Size of ArrayList: " + fruits.size());

System.out.println("----------------------------");

Vector<Integer> numbers = new Vector<>();
numbers.add(10);
numbers.add(20);
numbers.add(30);
System.out.println("Vector after adding: " + numbers);

numbers.remove(1);
System.out.println("Vector after removing index 1: " + numbers);

numbers.add(0, 5);
System.out.println("Vector after inserting at index 0: " + numbers);

System.out.println("Size of Vector: " + numbers.size());

System.out.println("----------------------------");

StringBuffer sb = new StringBuffer("Hello");
```

```
        System.out.println("Original StringBuffer: " + sb);

        sb.append(" World");
        System.out.println("After append: " + sb);

        sb.insert(5, ",");
        System.out.println("After insert: " + sb);

        sb.replace(0, 5, "Hi");
        System.out.println("After replace: " + sb);

        sb.reverse();
        System.out.println("After reverse: " + sb);

        sb.reverse();
        sb.delete(0, 2);
        System.out.println("After delete: " + sb);
    }
}
```

Output

```
ArrayList after adding: [Apple, Banana, Mango]
ArrayList after removing Banana: [Apple, Mango]
ArrayList after inserting Grapes: [Apple, Grapes, Mango]
Element at index 0: Apple
Size of ArrayList: 3
----------------------------
Vector after adding: [10, 20, 30]
Vector after removing index 1: [10, 30]
Vector after inserting at index 0: [5, 10, 30]
Size of Vector: 3
----------------------------
Original StringBuffer: Hello
After append: Hello World
After insert: Hello, World
After replace: Hi, World
After reverse: dlroW ,iH
After delete: , World

=== Code Execution Successful ===
```

Conclusion

This program demonstrates the use of ArrayList, Vector, and StringBuffer for dynamic data storage and manipulation in Java. ArrayList and Vector allow elements to be added, removed, and inserted at specific positions, growing dynamically unlike arrays. StringBuffer supports mutable string operations such as append, insert, replace, reverse, and delete, making it efficient for frequent string modifications compared to immutable Strings.